

Deep Learning IndabaX South Africa 2025

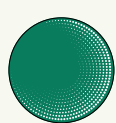
STELLENBOSCH, SOUTH AFRICA

7-11 JULY 2025

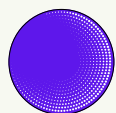
The Deep Learning IndabaX SA is the annual meeting of the South African machine Learning and AI community.

IndabaX, a branch of the pan-African movement *Deep Learning Indaba*, plays a pivotal role in expanding the **reach** and **impact** of **machine learning knowledge across Africa**.

- ✓ Spread **scientific knowledge** amongst researchers
- ✓ Showcase **cutting-edge applications** from industry partners
- ✓ Provide a **practical introduction** to deep learning for beginners
- ✓ Build a **community** of responsible AI practitioners



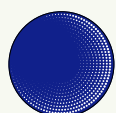
- PROGRAMME
ELEMENT



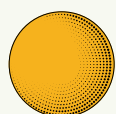
- WORKSHOP



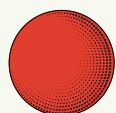
- DISCUSSION



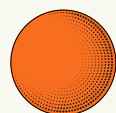
- KEYNOTE



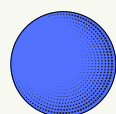
- EVENT



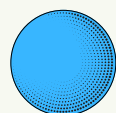
- FUNDAMENTALS



- STREAM 1



- STREAM 2



- HACKATHON

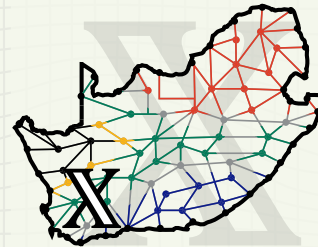
DAY 1

MONDAY

7 JULY '25

Deep Learning Indaba

South Africa 2025



08:00	Registration Continues throughout the day, and during each morning coffee <i>Civil Engineering Entrance Hall</i>			
08:30				
09:00		Soft Skill Monday Workshop: Maths for ML A friendly Introduction to the Mathematical Basics for Machine Learning <i>Labs: M2002</i>	Soft Skill Monday Workshop: Africa Comm (PhD level communication) Building Science Communication Capacity in Africa <i>S1018</i>	Soft Skill Monday Workshop: Scientific Communication Beyond the Poster: Practical Presentation Skills for Researchers <i>E1025</i>
09:30				
10:00				
10:30				
11:00	Debate <i>S1018</i>			
11:30	Lunch			
12:00				
12:30	Opening Address <i>S1018</i>			
13:00	Keynote Stuart Reid: The Unsung AI Scaling Law <i>S1018</i>			
13:30				
14:00	Welcoming - AI Fest with Nosible <i>S1018</i>			
14:30	Group Photo <i>Civil Engineering (outside)</i>			
15:00	Student Poster Session <i>Engineering Building Skywalk (3rd and 4th floors)</i>			
15:30				
16:00				
16:30	Music, Welcome Drinks <i>Civil Engineering Entrance Hall</i>			
17:00	Exhibition Booth Session <i>Civil Engineering Entrance Hall</i>			
17:30				
18:00				
18:30	Movie night: AlphaGo <i>Pulp Cinema, Neelsie</i>			
to late				

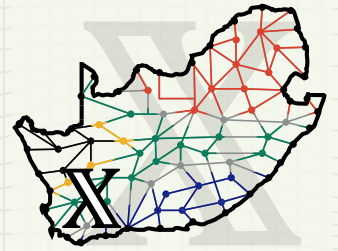
DAY 2

TUESDAY

8 JULY '25

Deep Learning Indaba

South Africa 2025



08:00	Morning coffee (late registration 8am - 8:30am)			Civil Engineering Entrance Hall
08:30				
09:00	Keynote			
09:30	Prof. Pius Owolawi: Edge AI for Agriculture - Towards a Cognitive Farming			S1018
10:00	Break (Speaker Intros start at 10:25am)			
10:30	Foundations of Machine Learning Prof. Mandla Gwetu	[Security] David de Villiers Overlooked and Undersecured: Prompt Injection Attacks on LLM Applications	[Ethics] Zara Schroeder Unconsented: The Gendered Cost of Generative AI in the Digital Age	
11:00		[Probabilistic models] Shane Josias Multimodal base distributions in conditional flow matching generative models	[Ethics] Stephen Middleton The Patent Landscape for Inventions relating to Artificial Intelligence	
11:30		[CV] Alaa Aldein Mohamed Suliman Ibrahim Transfer learning-based approach using new convolutional neural network classifier for steel surface defects classification	[Blockchain] Dr. Sebonkile Thaba From Supply Chains to Smart Solutions: Empowering Non-Technical Students with Blockchain Innovation	
	S1018	E1025	S2002	
12:00	Debate			
12:30				
13:00	Lunch			
13:30	Keynote			
14:00	Prof Marlene Holmner: Ethics in the Age of AI: Balancing Privacy, Accuracy, and Access			S1018
14:30	Break (coffee only) (Speaker Intros start at 14:55am)			
15:00	Foundations of Natural Language Processing Dr. Joan Byamugisha: Evolution of NLP Methods, and their Fit-for-purpose	[Algorithms] Bigomokero Antoine Bagula Artificial Learning: From Federated Learning to Diversity Learning	[NLP] Francois Meyer Improving Text Generation for Low-Resource South African Languages	
15:30		[Applications] Francois Naude Augmentation and Training Strategies for Improved Phase Arrival Picking in Microseismic Monitoring	[NLP] Stefan Strydom Pocket Polyglot Mzansi: Machine Translation for Local Languages on the Edge	
16:00		[Algorithms] Ahmed Mohammed Ahmed Grokking in neural networks and the role of regularisation [Prob. Models] Jimoh Saheed Tunji Evaluation of Wind Speed Regime Variability and Model Fit Performance Using Multi-Component Weibull Mixture Functions for Energy Potential Estimation		
	S1018	E1025	S2002	
16:30	Break & Snacks (Speaker Intros start at 16:55am)			
17:00	Flash Talks	[Zindi] Amy Bray Competitive Machine Learning to Improving your CV	Hackathon Practical With the Institute of Mine Seismology (5:00pm - 8pm) Whether you're just starting out or already have experience, join us for an exciting hands-on session where we tackle a real-world challenge straight from the mining industry.	
17:30				
18:00	Break			
18:30	Movie night: The Creator			
to late				
	Pulp Cinema, Neelsie		Labs: M2002, M3002	

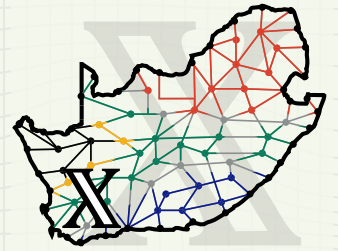
DAY 3

WEDNESDAY

9 JULY '25

Deep Learning Indaba

South Africa 2025

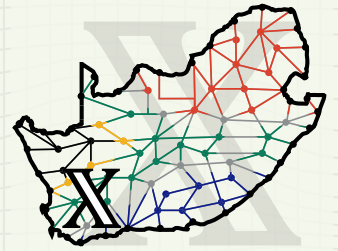


08:00	Morning coffee (late registration 8am - 8:30am)			
08:30				
09:00	Keynote			
09:30	Dr. Axel Tideman: Machine Learning at Takealot			S1018
10:00	Break (Speaker Intros start at 10:25am)			
10:30	Foundations of MLIP Jeroen Van Goey: Machine Learning Interatomic Potentials (MLIP) S1018	[MLOps] Stefan Dominicus Recommender Systems at Takealot.com	Naweed Khan The state of AI	Workshop Eyes in the sky: Applied Deep Learning with MATLAB and Minidrones - Optimum Solutions Labs: M2002
11:00		[MLOps] Jason Max Chalom Building out a sustainable and production ready AI software solution	[Compute] Luke Taylor Why AI is Power-Hungry - Lessons from the Brain and How We Can Fix It	
11:30		[RL] Samuel Oladejo ForecastRL: Neural Time Series Forecasting Integration with Risk-Aware Deep Reinforcement Learning for Financial Trading E1025	[KRR] Dr. Gavin Rens Symbolic Methods in Artificial Intelligence and Machine Learning S2002	
12:00	Debate: “I, Human Autonomy, and Power” by Dr Hossein Masoumi Karakani			S1018 Labs: M2002
12:30	Lunch			
13:00				
13:30	Workshop Recommender Systems Workshop - Takealot S1018	Workshop AI Safety - AI Safety Cape Town E1025	Workshop IBM Skills Build Labs: M3002	Workshop Eyes in the sky: Applied Deep Learning with MATLAB and Minidrones - Optimum Solutions Labs: M2002
14:00				
14:30	Break (coffee only) (Speaker Intros start at 14:55am)			
15:00	Workshop Recommender Systems Workshop - Takealot S1018	Workshop AI Safety - AI Safety Cape Town E1025	Workshop IBM Skills Build Labs: M3002	Workshop Eyes in the sky: Applied Deep Learning with MATLAB and Minidrones - Optimum Solutions Labs: M2002
15:30				
16:00				
16:30	Break & Snacks (Speaker Intros start at 16:55am)			
17:00	Flash Talks S1018		Radixtrie Hackathon (5:00pm - Midnight) Ever been curious to apply ML algorithms to Databases? Now is your chance! Take part in this challenge and win. Labs: M2002	Instadeep Hackathon (5:00pm - Midnight) Join us for a formal hackathon session with Instadeep focusing on ML for Interatomic Potentials (MLIP). Get an introduction to the field and prepare for the hackathon challenge ahead. Join us to win some cool prizes! Labs: M3002
17:30				
18:00	Break			
18:30	Movie night Pulp Cinema, Neelsie			
to late				

DAY 4
THURSDAY
10 JULY '25

Deep Learning Indaba

South Africa 2025



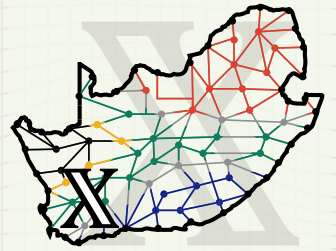
08:00	Morning coffee (late registration 8am - 8:30am)		
08:30			
09:00	Keynote		
09:30	Prof. Francesco Petruccione: Quantum Foundations - From Qubits to Intelligent Algorithms		S1018
10:00	Break (Speaker Intros start at 10:25am)		
10:30	Foundations of Data Literacy Dr Hossein Masoumi Karakani: Data Culture Rising - Literacy for the Next Generation S1018	Workshop Quantum Machine Learning 4 Africa - IBM Research Africa Labs: M2002	Workshop Building AI Agentic Workflows from Scratch: A Hands-On Tutorial on RAG, Text2SQL & Tools Labs: M3002
11:00			
11:30			
12:00	Lunch		
12:30			
13:00	[Compute] Werner Janse van Rensburg Introduction to NICIS: Your Compute (CHPC), Data (DIRISA) and Network (SANReN) Infrastructure Services Partner	Workshop Quantum Machine Learning 4 Africa - IBM Research Africa Labs: M2002	Workshop Building AI Agentic Workflows from Scratch: A Hands-On Tutorial on RAG, Text2SQL & Tools Labs: M3002
13:30	[NLP] Avi Moodley NLP Solutions for Practical Industry Use Cases S1018		
14:00	Break (coffee only) (Speaker Intros start at 14:25am)		
14:30	[Ethics] Karen Ann Walstra AI School Policies: An Urgent Need [Ethics] Gabriela Penelopé Carolus Ground Truth: Blending Clinical Evidence and Unstructured Deep Learning for a Culturally Responsive AI Health Innovation	Workshop Quantum Machine Learning 4 Africa - IBM Research Africa Labs: M2002	Workshop Building AI Agentic Workflows from Scratch: A Hands-On Tutorial on RAG, Text2SQL & Tools Labs: M3002
15:00	[Health] Craig Parker Beyond Weather Warnings: Machine Learning Reveals the Hidden Biology of Climate Health Vulnerability [Applications] Dylan Geldenhuys Building a Digital Twin for Telco: Optimizing Customer Satisfaction with Causal ML - Simulating interventions, predicting customer impact, and guiding network decisions at scale S1018		
15:30	Break & Snacks (Speaker Intros start at 15:55am)		
16:00	Panel Discussion: 70 years of AI		
16:30			
17:00	Prize Giving and Closing Ceremony		S1018
17:30	Travel		
18:00	Closing Event		
18:30			
to late			

Dorp, Stellensbosch

Dorp, Stellensbosch

DAY 5
FRIDAY
11 JULY '25

Deep Learning Indaba
South Africa 2025



REFLECT, REFINE, AND RELAX

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NEELSIE CAFETERIA

We've come to the final day of the IndabaX. After a packed week of learning, sharing, and building community, take time to slow down, wrap up, and enjoy Stellenbosch with new friends before heading home.

08:00		
08:30		
09:00	Informal Focus Group & Week Wrap-Up We never get enough time to chat! Grab a coffee and come chat with the team. We'd love to hear how the week went for you! What worked, what could be better, and what you're taking home. It's informal and open to all.	
09:30	CV Clinics & Soft Skills Wrap-up One-on-one time with organisers and IBM mentors. If you've been working on a CV, project pitch, or presentation, now's your chance to get final input before you leave.	"Walk & Wonder" Stellenbosch Tour Stretch your legs and enjoy the area. Choose from a short scenic hike, a visit to the botanical gardens, or a relaxed museum trip. Perfect if you've got some time before travelling.
10:00		
10:30		
11:00	Explore Stellenbosch Use the rest of the day to explore with new friends. The Neelsie and town centre are great for lunch, gift shopping, or grabbing a last coffee. If you're around a bit longer, the nearby vineyards are easy to reach and make for a chilled afternoon.	
11:30		
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